WEEK 1 · DAY 4 · OVERVIEW

Regression Assumptions and Log Spreads



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ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Day 3's kept model, 17 rating steps and years to maturity, misses a typical unseen bond by 35.4 bp.

statsmodels puts AAA's step at plus or minus 36.7 bp, while 500 refits on resampled bonds move it by only 5.6 bp.

Before anyone quotes a step, the desk wants every assumption under the model checked, and a fix for any that fails.

The bonds and issuers are invented for this course.

OBJECTIVES

Check linearity, constant variance, normality and independence, each with a chart and a number.

Find why the bp model fails two of them, refit on the log of OAS, and check again.

Bring the log model back to bp, read a rating step as a percent, repair the error bars, and decide what to do with six distressed bonds.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 4

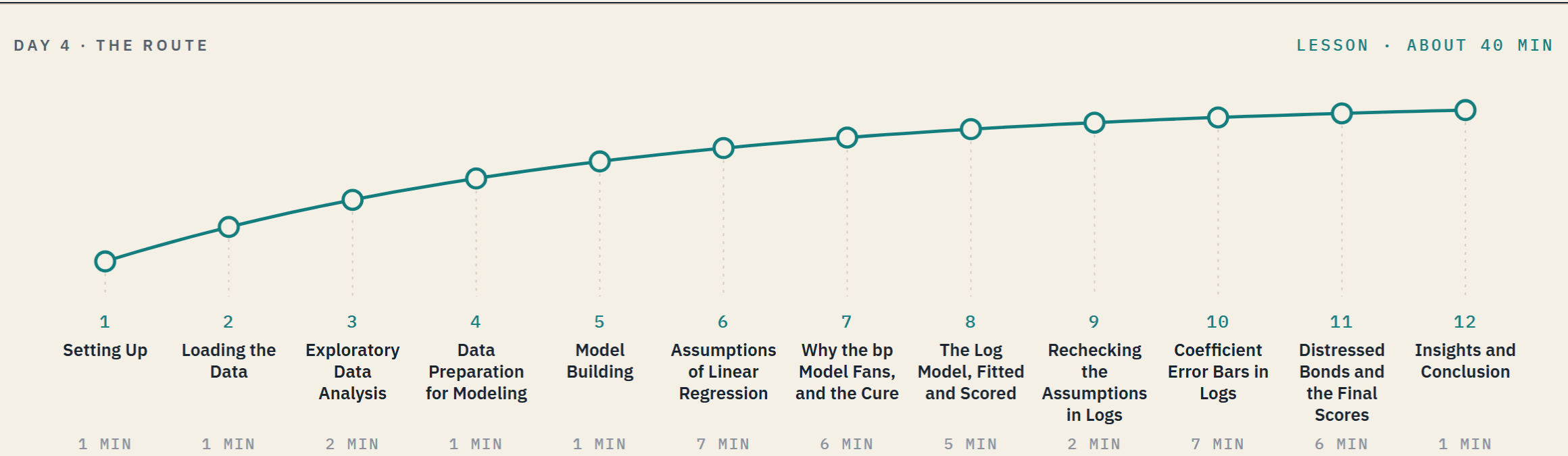


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 40 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SEED AND THE DATA

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.stats.diagnostic import het_breuschpagan
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
sns.set_theme(style="whitegrid")
sns.set_palette("colorblind")
plt.rcParams["figure.figsize"] = (10, 5)
pd.set_option("display.float_format", "{:,.3f}".format)
```

Loading the Data

THE DATA DICTIONARY

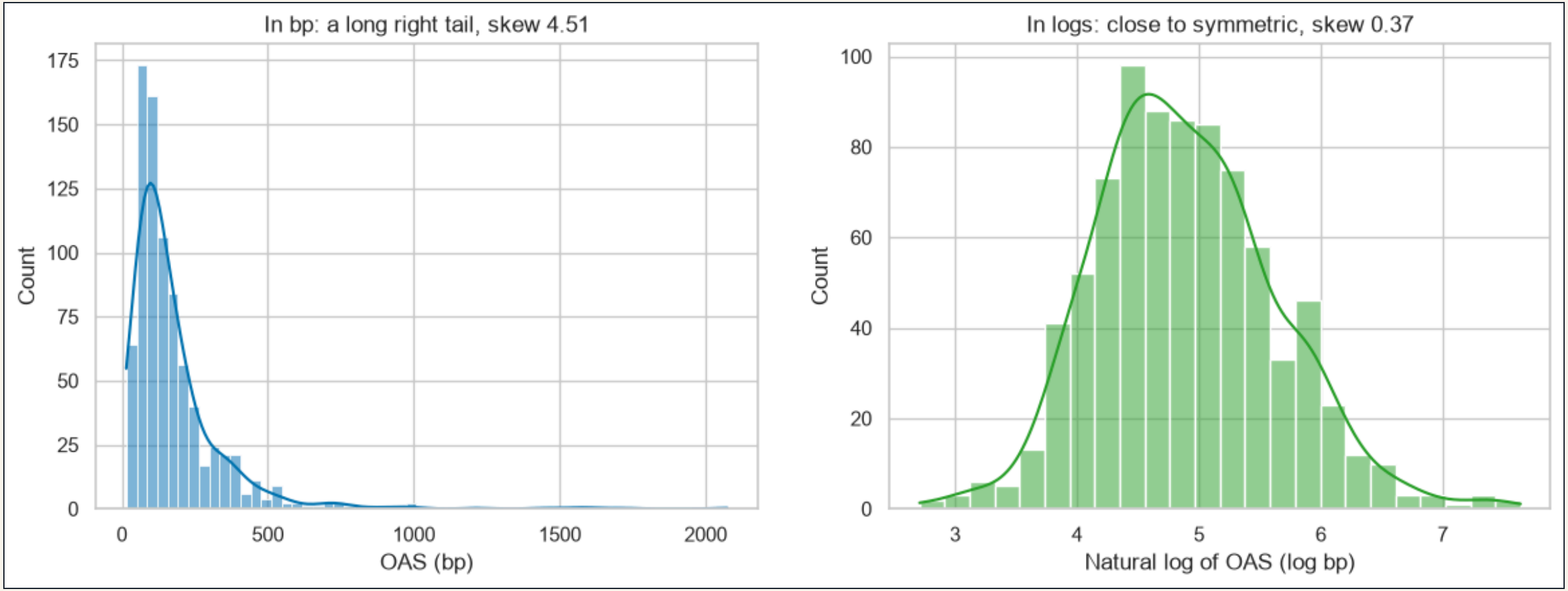
- **`cusip`**: the bond's identifier
- **`ticker`**: the issuer's short code, shared by all of its bonds
- **`issuer`**: the issuer's name
- **`rating`**: credit rating, AAA to CCC
- **`oas`**: option-adjusted spread, bp, **the target**
- **`maturity`** and **`as_of_date`**: dates, from which years to maturity is built

READING THE BONDS

```
bonds = pd.read_csv(DATA + "benchmark.csv")
ratings = pd.read_csv(DATA + "ratings.csv")
data = bonds.merge(ratings[["rating", "score", "grade"]], on="rating")
data["years_to_maturity"] = (pd.to_datetime(data["maturity"]) - pd.to_datetime(data["as_of_date"])).dt.days / 365.25
```

Check the spreads in bp and in logs.

Exploratory Data Analysis



Spreads are lopsided in bp and close to symmetric in logs. Hold that thought, because first we test the bp model as Day 3 left it.

Rebuild Day 3's columns and split.

Data Preparation for Modeling

DAY 3'S COLUMNS AND SPLIT

```
rating_order = ratings["rating"]
rating_category = pd.Categorical(data["rating"], categories=rating_order)
rating_dummies = pd.get_dummies(rating_category, prefix="rating", dtype=float).drop(columns="rating_A")
X = pd.concat([rating_dummies, data["years_to_maturity"]], axis=1)
y = data["oas"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=SEED)
print("Training bonds:", len(X_train), " Test bonds:", len(X_test))
```

```
Training bonds: 615  Test bonds: 206
```

The same columns, bonds and split as Day 3's kept model.

Refit Day 3's model in statsmodels.

Model Building

REFIT IN STATSMODELS

```
bp_model = sm.OLS(y_train, sm.add_constant(X_train)).fit()
print("R-squared:", round(bp_model.rsquared, 4))
```

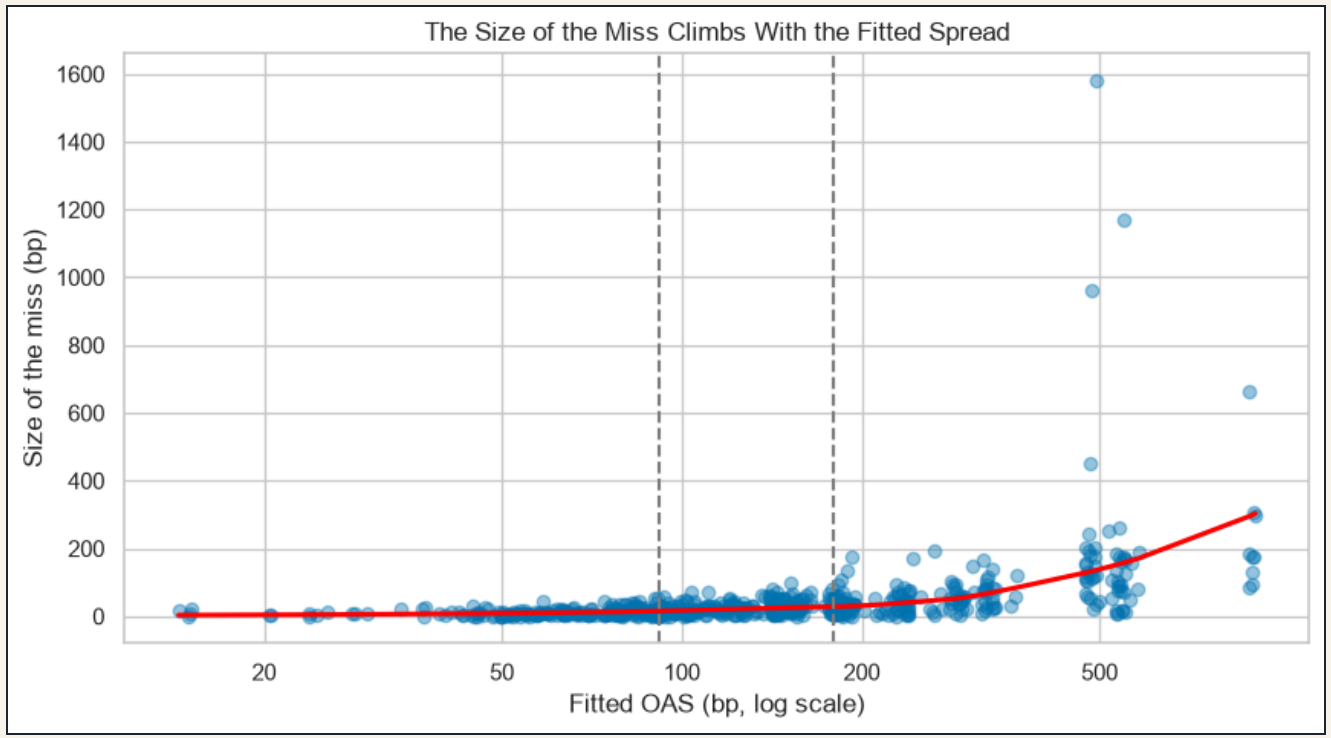
```
R-squared: 0.6746
```

FITTED VALUES AND MISSES

```
bp_checks = pd.DataFrame({"fitted": bp_model.fittedvalues, "residual": bp_model.resid})
bp_checks.head()
```

	FITTED	RESIDUAL
712	50.746	-1.746
181	77.864	-18.864
778	28.038	-6.038
716	144.360	72.640
636	118.649	9.351

Assumptions of Linear Regression



NOISE IN EACH THIRD

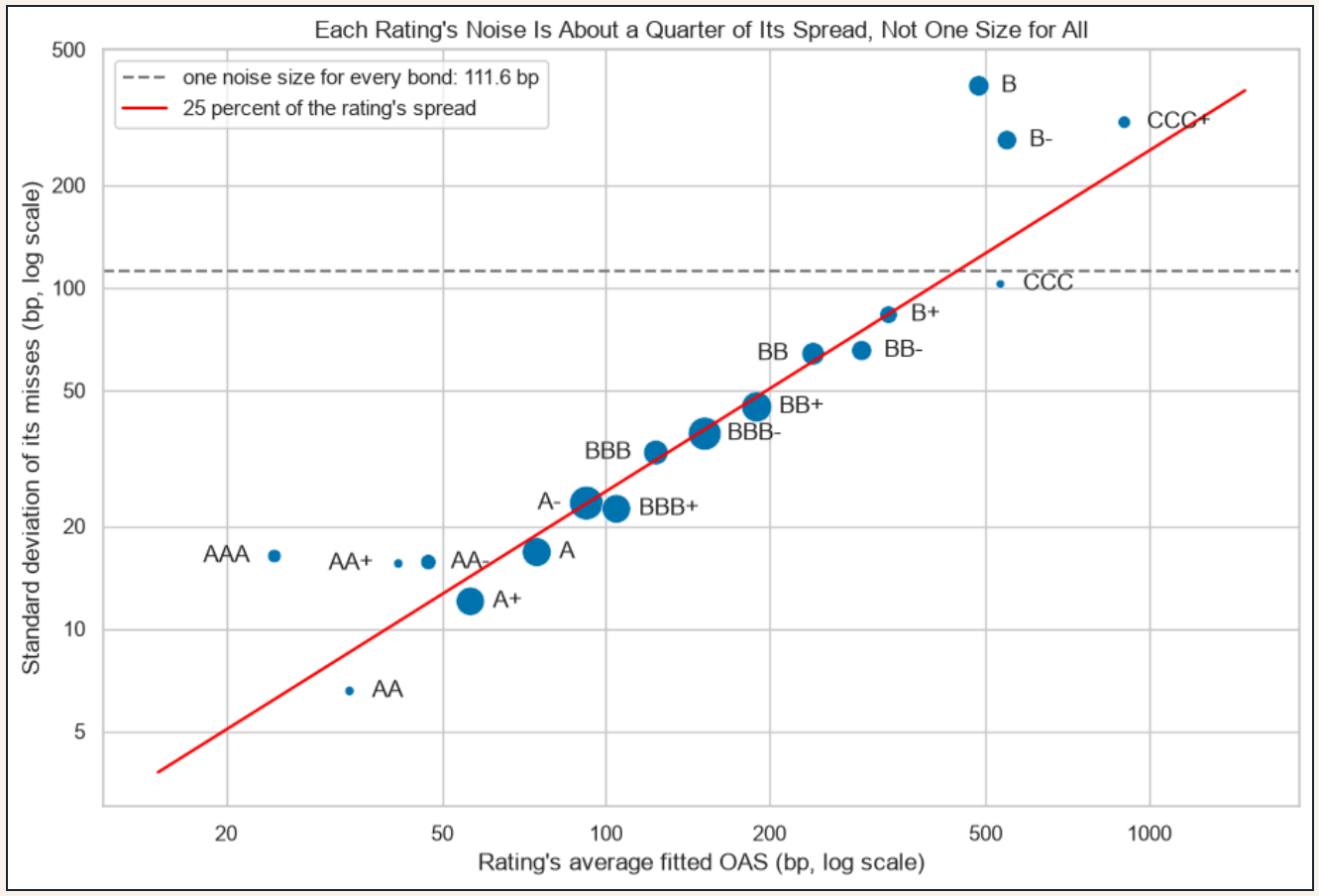
```
bottom    15.866
middle    31.212
top      187.483
dtype: float64
Top over bottom: 11.8
```

BREUSCH-PAGAN

```
Breusch-Pagan statistic: 47.6  p-value: 0.00017
```

Shape passes, while constant variance and normality fail badly. Next, why a bp model fans.

Why the bp Model Fans, and the Cure



THE RECIPE

A bond's spread is its rating's base times a maturity factor, times its issuer's factor, times a factor of its own.

THE FIRST RATINGS

RATING	AAA	AA+	AA	AA-
level	24.545	41.500	33.750	47.133
noise	16.320	15.519	6.561	15.667
bonds	11.000	4.000	4.000	15.000
noise_share	0.665	0.374	0.194	0.332

From A+ to B+, ratings with 20 to 80 training bonds each, the noise is 21 to 27 percent of the spread.

The Log Model, Fitted and Scored

FIT ON LOG OAS

```
log_model = sm.OLS(np.log(y_train), sm.add_constant(X_train)).fit()
print("R-squared, in logs:", round(log_model.rsquared, 4))
```

```
R-squared, in logs: 0.8907
```

ONE TEST BOND'S MISS UNDER BOTH MODELS, IN BP

```
test_misses[data.loc[X_test.index, "cusip"] == "99038MAC3"]
```

	BP MODEL	LOG MODEL
196	714.790	785.771

The test bonds, in bp, decide.

Rechecking the Assumptions in Logs

DO BONDS OF ONE ISSUER MISS TOGETHER?

```
issuer = data.loc[X_train.index, "ticker"]
shuffled = np.random.default_rng(SEED).permutation(issuer.to_numpy())
for name, labels in [("real issuers", issuer), ("shuffled", shuffled)]:
    average = log_model.resid.groupby(labels).transform("mean")
    print(name, round((average ** 2).sum() / (log_model.resid ** 2).sum() * 100, 1), "percent")
```

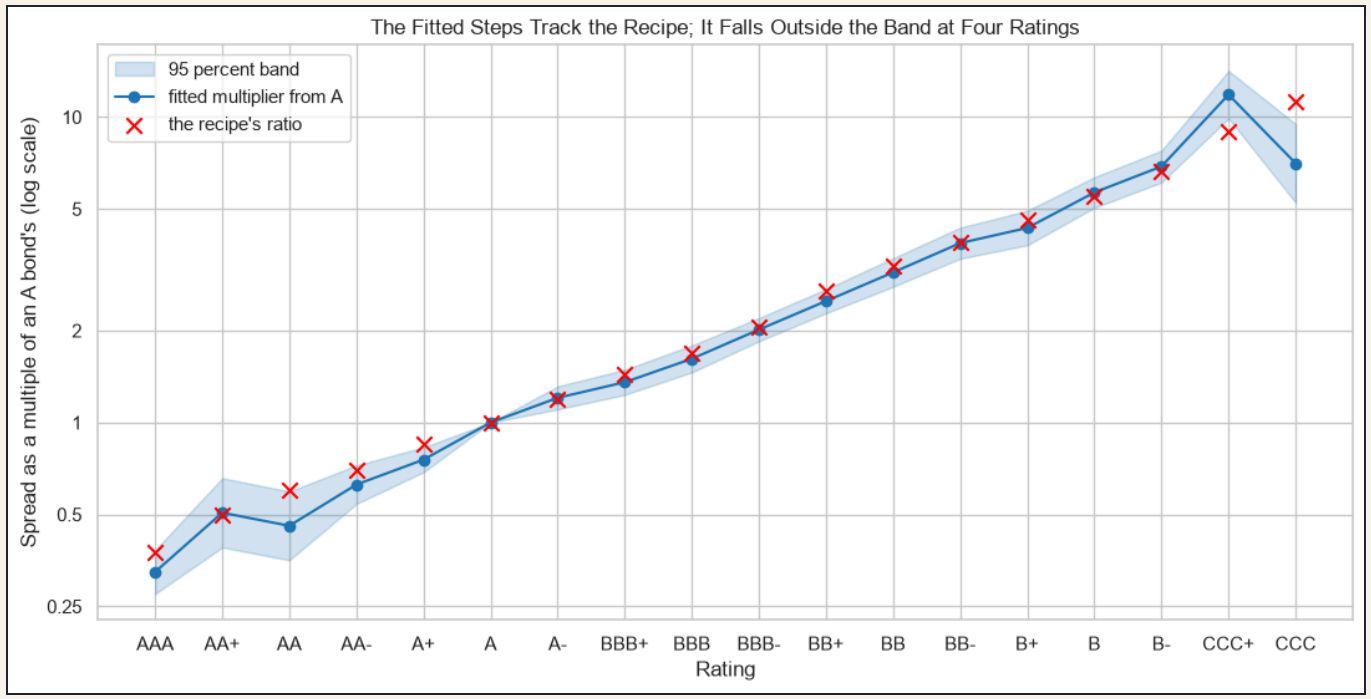
```
real issuers 42.8 percent
shuffled 26.0 percent
```

THE THREE TESTS IN LOGS

```
Breusch-Pagan p-value: 0.00041102894492692 53
Shapiro-Wilk p-value: 3.228611440549125e-11
Jarque-Bera p-value: 2.58316088875301e-88
```

42.8 percent with real issuers against 26.0 percent shuffled.

Coefficients and Error Bars in Logs

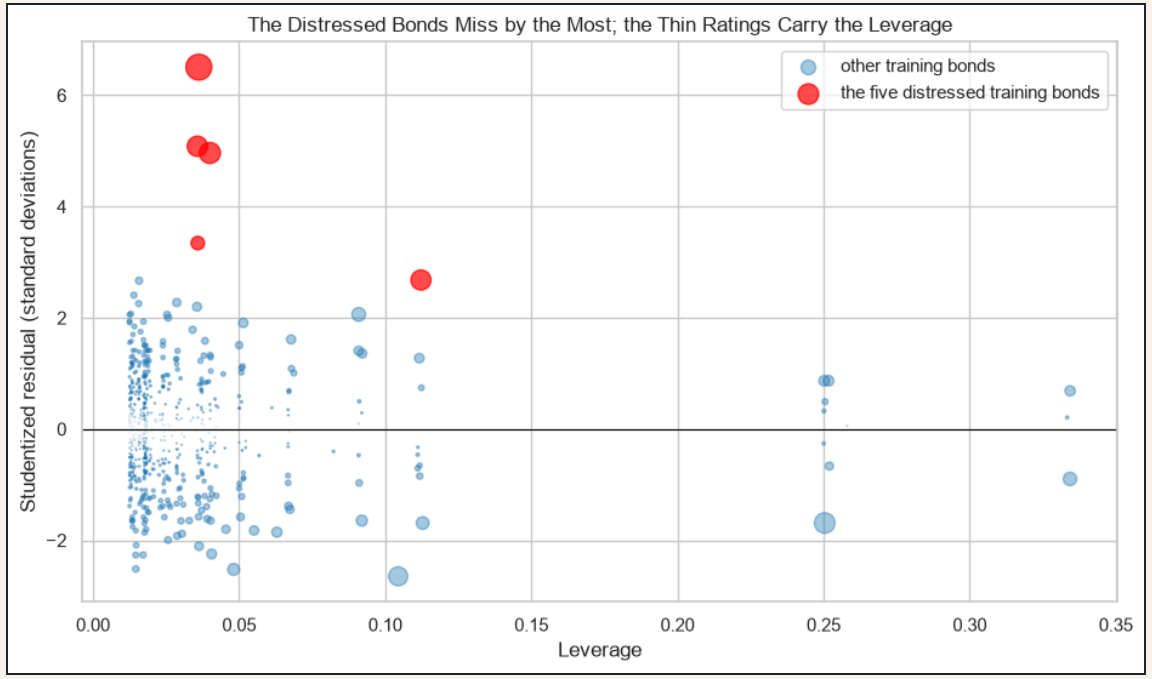


A STEP AS A PERCENT FROM A

	0	1	PERCENT
rating_AAA	-72.489	-61.766	-67.568
rating_BB	178.526	245.298	210.120
years_to_maturity	0.662	1.275	0.968

Holding years fixed, a BB bond's spread sits 210.1 percent above an A bond's, with an interval of 178.5 to 245.3.

Distressed Bonds and the Final Scores

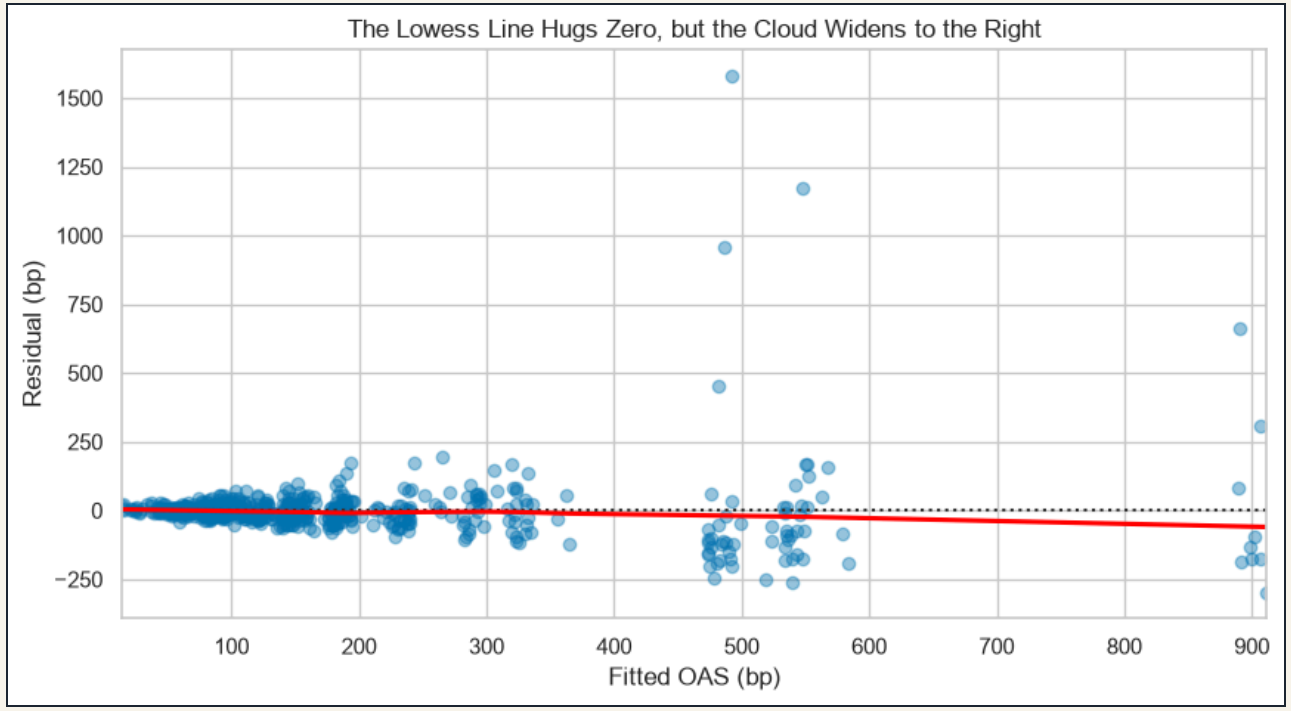


THE SIX MOST LIKELY DISTRESSED BONDS

	CUSIP	ISSUER	RATING	OAS	TRAINING
386	99072UAF5	Zephane Utilities	B	2074	True
47	99009JAA9	Calderby Retail	B	1446	True
789	99145PAB8	Praximer Energy	B-	1718	True
533	99098UAF5	Elvundra Foods	B	933	True
365	99069RAC4	Feskovar Devices	CCC+	1551	True
196	99038MAC3	Praxiskan Finance	CCC+	1605	False

Five training bonds carry the largest misses the log model leaves.

Insights and Conclusion



THE AVERAGE MISS IN EACH FIFTH

```
fitted
(14.39, 77.166]       1.606
(77.166, 101.923]     0.244
(101.923, 151.771]   -2.496
(151.771, 239.312]   -3.702
(239.312, 911.763]    4.348
Name: residual, dtype: float64
```

Day 5 starts from the log target and these columns.

You Can Now

- Check linearity, constant variance, normality and independence, each with a chart and a number.
- Read a lowess line, a scale-location plot and a Q-Q plot.
- Fit a model on log OAS, bring it back to bp with **`np.exp`**, and say why **exp** gives the middle bond.
- Read a log coefficient as a percent.
- Mend error bars with HC3 or with logs, and find the bonds that pull hardest on a fit.

NEXT
WEEK 1 · DAY 5

Case Study: Ridge, Lasso, and the Spread Cross-Section

On Day 5 you put all of it to work on a case study.